observe that nuclear reaction formulas and energies refer to single atoms (or molecules), while chemical reactions and equations refer to number of moles; we have:

$$1\ \text{eV/molecule} = 1.6022 \times 10^{-19} \times 6.0221 \times 10^{23} = 96.48\ \text{kJ mole}^{-1}$$
$$= 3.8268 \times 10^{-20} \times 6.0221 \times 10^{23} = 23.045\ \text{kcal mole}^{-1} \quad (4.8)$$

Thus, the formation of deuterium from a neutron and a hydrogen atom would lead to the liberation of 214.6×10^6 kJ (51.3×10^6 kcal) for each mole of deuterium atoms formed. By comparison, then, the nuclear reaction leading to the formation of deuterium is approximately half a million times more energetic than the chemical reaction leading to formation of CO_2.

It is not common practice to use mole quantities in considering nuclear reactions as the number of individual reactions under laboratory conditions is well below 6.022×10^{23}. Therefore, in nuclear science one uses the energy and mass changes involved in the reaction of individual particles and nuclei.

4.4. BINDING ENERGY

The energy liberated in the formation of CO_2 from the elements, the heat of formation, is a measure of the stability of the CO_2 molecule. The larger the heat of formation the more stable the molecule since the more energy is required to decompose the molecule into its component atoms. Similarly, the energy liberated in the formation of a nucleus from its component nucleons is a measure of the stability of that nucleus. This energy is known as the *binding energy* (E_B) and has the same significance in nuclear science as the heat of formation has in chemical thermodynamics. We have seen that the binding energy of deuterium is 2.22 MeV. The $^4_2$He nucleus is composed of 2 neutrons and 2 protons. The measured mass of the ^{4}He atom is 4.002 603 u. The mass defect is:

$$\Delta M_{He} = M_{He} - 2\,\mathbf{M}_H - 2\,\mathbf{M}_n = 4.002603 - 2 \times 1.007825 - 2 \times 1.008665$$
$$= -0.030377\ \text{u} \quad (4.9)$$

According to eqn. (4.6) the binding energy, E_B, between the nucleons in a nucleus follows the simple relation[2]

$$E_B(\text{MeV}) = -931.5\ \Delta M_A(\text{u}) \quad (4.10)$$

Thus the binding energy for ^{4}He is 28.3 MeV. It is quite unlikely that 2 neutrons and 2 protons would ever collide simultaneously to form a ^{4}He nucleus; nevertheless, this calculation is useful because it indicates that to break ^{4}He into its basic component nucleons would require at least 28.3 MeV.

[2] The minus sign indicates that the loss of mass in formation of a nucleus releases energy.

A better indication of the relative stability of nuclei is obtained when the binding energy is divided by the total number of nucleons to give the average binding energy per nucleon, E_B/A. For ^{4}He the value of E_B/A is 28.3/4 or 7.1 MeV, whereas for ^{2}H it is 1.11 for the bond between the two nucleons. Clearly, the ^{4}He nucleus is considerably more stable than the ^{2}H nucleus. For most nuclei the values of E_B/A vary in the rather narrow range 5–8 MeV. To a first approximation, therefore, E_B/A is relatively constant which means that the total nuclear binding energy is roughly proportional to the total number of nucleons in the nucleus.

Figure 4.3 shows that the E_B/A values increase with increasing mass number up to a maximum around mass number 60 and then decrease. Therefore the nuclei with mass numbers in the region of 60, i.e. nickel, iron, etc., are the most stable. Also in this Figure we see that certain numbers of neutrons and protons form especially stable configurations – this effect is observed as small humps on the curve.

If two nuclides can be caused to react so as to form a new nucleus whose E_B/A value is larger than that of the reacting species, obviously a certain amount of binding energy would be released. The process which is called *fusion* is "exothermic" only for the nuclides of mass number below 60. As an example, we can choose the reaction

$$^{20}_{10}\mathrm{Ne} + {}^{20}_{10}\mathrm{Ne} \rightarrow {}^{40}_{20}\mathrm{Ca} \tag{4.11}$$

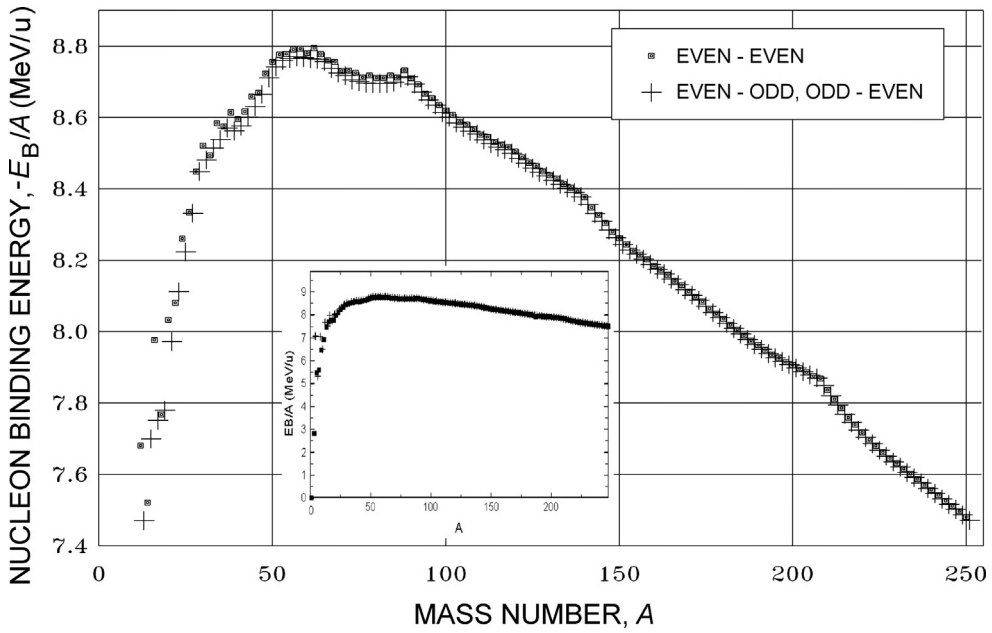


Figure 4.3 Binding energy per nucleon (E_B/A) for the most stable isobars as function of mass number (A).

From Figure 4.3 we estimate that E_B/A for neon is about 8.0 MeV and for calcium about 8.6 MeV. Therefore, in the 2 neon nuclei $2 \times 20 \times 8.0 = 320$ MeV are involved in the binding energy, while $40 \times 8.6 = 344$ MeV binding energy are involved in the calcium nucleus. When 2 neon nuclei react to form the calcium nucleus the difference in the total binding energy of reactants and products is released; the estimate gives $344-320 = 24$ MeV; a calculation using measured masses gives 20.75 MeV. It is also evident that the greater gain in fusion energy is in the combination of lighter elements where the derivative of the E_B/A curve is greater, e.g. $^{2}H + {}^{2}H \rightarrow {}^{4}He$ liberates 23.85 MeV per atom of ^{4}He produced.

Figure 4.3 also shows that a similar release of binding energy can be obtained if the elements with mass numbers greater than 60 are split into lighter nuclides with higher E_B/A values. Such a process, whereby a nucleus is split into two smaller nuclides, is known as *fission*. An example of such a fission process is the reaction

$$^{236}_{92}U \rightarrow {}^{140}_{54}Xe + {}^{93}_{38}Sr + 3n \tag{4.12}$$

The binding energy per nucleon for the uranium nucleus is 7.6 MeV, while those for the ^{140}Xe and ^{93}Sr are 8.4 and 8.7 MeV respectively. The amount of energy released in this fission reaction is approximately $140 \times 8.4 + 93 \times 8.7 - 236 \times 7.6 = 191.5$ MeV for each uranium fission.

4.5. NUCLEAR RADIUS

Rutherford showed by his scattering experiments that the nucleus occupies a very small portion of the total volume of the atom. Roughly, the radii of nuclei vary from 1/10 000 to 1/100 000 of the radii of atoms. While atomic sizes are of the order of 100 pm (10^{-10} m), the common unit of nuclear size is the femtometer (1 fm = 10^{-15} m), sometimes referred to as 1 Fermi.

Experiments designed to study the size of nuclei indicate that the volumes of nuclei (V_n) are directly proportional to the total number of nucleons present, i.e.

$$V_n \propto A \tag{4.13}$$

Since for a sphere $V \propto r^3$, where r is the radius of the sphere, for a spherical nucleus $r^3 \propto A$, or $r \propto A^{1/3}$. Using r_o as the proportionality constant

$$r = r_o A^{1/3} \tag{4.14}$$

The implications of this is that the nucleus is composed of nucleons packed closely together with a constant density (about 0.2 nucleons fm^{-3}) from the center to the edge of the nucleus. This constant density model of the nucleus as been shown to be not completely correct, however. By bombarding nuclei with very high energy electrons or protons (up to $\geq$1 GeV) and measuring the scattering angle and particle energy, the